

Alma Mater Studiorum · Università di Bologna

Department of Computer Science and Engineering
Master Degree in Artificial Intelligence

Artificial Intelligence in Industry Project

Forecasting Median House Prices

Luca Corradini Zini

Project Objective

- Predict monthly median house prices using historical property sales data
- Data source: [Kaggle property sales time series dataset](#)
- Understand which factors drive price dynamics over time
- Compare multiple forecasting approaches:
 - Random Forest
 - LSTM Neural Networks
 - SARIMAX Time Series Models
- Investigate unexpected model behaviour and improve the feature space

Dataset

- Data source: Kaggle property sales dataset
- Time span: 2007 – 2019
- Each observation corresponds to a property sale
- Key variables:
 - sale date
 - property price
 - property type (house / unit)
 - number of bedrooms
 - postcode
- Goal: predict a **monthly time series of housing prices**

Project outline

- Problem & dataset
- Extensive EDA
- Data issues and preprocessing (missing values and outliers)
- Feature engineering
- ML models evaluation (RF, LSTM, SARIMAX)
- Feature importance analysis
- Observations of results and hypothesis exploration

Dataset

Index	datesold	postcode	price	propertyType	bedrooms
0	2007-02-07	2607	525000	house	4
1	2007-02-27	2906	290000	house	3
2	2007-03-07	2905	328000	house	3
3	2007-03-09	2905	380000	house	4
4	2007-03-21	2906	310000	house	3

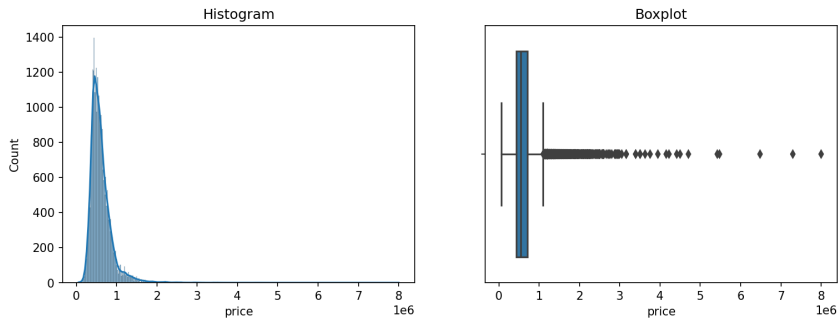
- Each row corresponds to a transaction
- Dataset contains thousands of observations
- Time information allows the construction of temporal features

Exploratory Data Analysis

Goals of the EDA:

- Understand distributions
- Identify missing values or anomalies
- Study relationships between features
- Identify potential predictive variables

Distribution of Property Prices



- Prices are strongly right-skewed
- Presence of very expensive properties
- Outliers may affect machine learning models

Distribution of property prices

```
avg_price_by_type = df.groupby('type')['price']. \
    mean().sort_values(ascending=False)
```


Missing values

The entries where

Outlier Handling

We explored different strategies:

- Keep data unchanged
- Remove extreme values using IQR rule
- Apply logarithmic transformation

These approaches were tested during model evaluation.

Average Price by Property Type

- Houses tend to be more expensive than units
- Property type is therefore an important explanatory feature

Average Price by Number of Bedrooms

- Moderate positive relationship
- Larger properties generally sell at higher prices

Feature Correlation

- Bedrooms show moderate correlation with price
- Suggests structural characteristics of houses affect price

From Transactions to Time Series

Individual sales were aggregated into monthly statistics:

- median price
- number of sales
- average bedrooms
- ratio of houses vs units

This creates a **structured monthly dataset** suitable for forecasting.

Monthly Price Time Series

- Clear long-term upward trend
- Potential seasonal fluctuations

Seasonal Decomposition

- Trend component shows long-term price increase
- Seasonal component suggests annual cycles
- Residual captures irregular fluctuations

Feature Engineering

Additional temporal features were introduced:

- Month encoded as cyclical variables
 - month_sin
 - month_cos
- Scaled year variable representing long-term trend

This helps machine learning models capture seasonality.

Modeling Approaches

Three families of models were explored:

- Random Forest Regressor
- LSTM Neural Networks
- SARIMAX Time Series Model

Each model captures temporal structure differently.

Random Forest Model

Key predictive variables:

- number of sales
- average bedrooms
- house ratio

Feature Importance

Most influential predictors:

- sales volume
- property composition
- housing characteristics

LSTM Neural Network

- Sequential neural network
- Captures temporal dependencies
- Uses previous months as input sequence

SARIMAX Model

- Traditional statistical time series model
- Handles trend and seasonality explicitly
- Allows additional exogenous variables

Unexpected Observation

Across models we noticed a systematic issue:

- Predictions tended to be **lower than real prices**
- Particularly in recent years
- Models captured trends but **underestimated price growth**

This raised an important question:

Why are prices increasing faster than predicted?

Hypothesis 1: Change in Property Types

Possible explanation:

- Houses are more expensive than units
- Perhaps recent years saw more houses sold

Result:

- No major structural change detected

Hypothesis 2: Geographic Shift

Maybe buyers increasingly purchased homes in expensive areas.
However:

- Premium postcode share remained relatively stable
- Does not fully explain price increase

Remaining Puzzle

Key empirical observation:

- Sales volume decreased in later years
- Yet prices continued rising

This contradicts simple supply-demand intuition.

Therefore:

Something outside the dataset may be influencing prices

Introducing Macroeconomic Variables

New hypothesis:

- Housing markets are influenced by macroeconomic conditions
- Important candidates:
 - mortgage interest rates
 - central bank interest rates
 - credit availability

These variables were integrated as **exogenous features**.

Macroeconomic Effects

Observation:

- Falling mortgage rates increase affordability
- Can drive price growth even when demand falls

Improved Model Performance

Including macro variables allows models to:

- better capture long-term dynamics
- explain rising prices despite falling sales
- reduce systematic underprediction

Key Takeaways

- Housing prices show strong temporal structure
- Sales volume and property characteristics are important predictors
- However, **micro-level data alone is insufficient**
- Macroeconomic variables play a crucial role in price dynamics

Conclusion

- Machine learning models can effectively forecast housing prices
- Understanding model errors can reveal deeper insights
- Integrating macroeconomic indicators significantly improves interpretation

Future work:

- incorporate additional economic indicators
- explore spatial models
- test alternative deep learning architectures

References

- Kaggle Housing Sales Dataset
- Housing Market Forecasting Literature
- Time Series Forecasting with Decomposition